

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **agentic workflows** orchestrating **LLM agents**, **OpenAI function calling**, and **MCP clients** using **Python** and **Node.js** backends integrated with **REST APIs** for production-grade reliability and SLA compliance, reducing manual workflow steps by **40%** across enterprise integrations.
- Built user-facing features in **React** and **Angular** with **TypeScript**, implementing **reusable UI components**, **service layers**, and **state management** for agent invocation, status visualization, and error handling; developed **Jest** unit tests and **Cypress** e2e tests achieving **85%+ code coverage**.
- Engineered **prompt engineering** patterns, **tool schema definitions**, **function calling orchestration**, and **context window management** for **Claude** and **OpenAI** models; implemented **memory and retrieval strategies** with vector embeddings and **RAG pipelines** to reduce hallucination by **35%** and improve agent reliability.
- Integrated **GitHub Actions**, **GitHub APIs**, **PR comment bots**, and **Cline AI coding assistant workflows** with agentic systems; implemented **OpenAI specification** contracts for tool definitions and function orchestration, enabling automated developer feedback loops across **200+ repositories**.
- Shipped production code with **CI/CD pipelines** (**Jenkins**, **Git**), **observability** via **OpenTelemetry** and structured logging for agent runs and UI interactions; enforced **input validation**, **secrets management**, **rate limiting**, and **least-privilege token patterns** for agent security and compliance.
- Authored **architecture documentation**, **OpenAI and MCP specification contracts**, **prompt versioning guidelines**, and **developer SDKs** for internal agent invocation; designed **prompt evaluation frameworks** and **golden test sets** for experiment validation, enabling **5+ cross-functional teams** to ship agent features independently.

SKILLS

AI & LLM Agent Frameworks – LLM agents, Claude models, OpenAI API, function calling, MCP (Model Context Protocol) clients/servers, prompt engineering, tool schemas, RAG pipelines, vector embeddings, context window management, memory strategies, prompt versioning, evaluation frameworks.

Frontend & Web Development – React, Angular, AngularJS, TypeScript, JavaScript, REST APIs, GraphQL, reusable UI components, design systems, accessibility standards, Jest, Cypress, linting, responsive design.

Backend & Agent Integration – Node.js, Python, Flask, FastAPI, agentic workflows, GitHub Actions, GitHub APIs, Cline workflows, tool orchestration, function calling patterns, error handling, retry logic, guardrails.

Infrastructure & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD (Jenkins, Git), Linux, shell scripting, observability (OpenTelemetry, structured logging), security (input validation, secrets management, rate limiting, least-privilege access).

Data & Backend Services – PostgreSQL, MySQL, Oracle, SQL, Python scripting, data pipelines, ETL, Spring Boot, REST API development, cross-system integrations, database schema design.

Domain & Compliance – Banking/payments operations, SWIFT messaging, ISO 20022, SEPA payments, PSD2, GDPR, regulatory compliance, SLA management, incident response, production support, change-control management.

Analytics & Visualization – Power BI, Tableau, DAX, Power Query, data modelling, data warehousing, statistical analysis, machine learning (scikit-learn, supervised/unsupervised learning).

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

AI Agent Developer Workflow Platform – GitHub-Integrated Agent Orchestration Suite (HCL Technologies)

- Architected and shipped an end-to-end **agentic workflow platform** enabling developers to invoke **Claude and OpenAI LLM agents** directly from **GitHub Pull Requests** via **PR comment bots** and **GitHub Actions**; integrated **Cline AI coding assistant workflows** with custom **MCP servers** and **OpenAI function calling** to automate code review, refactoring suggestions, and test generation.
- Designed **React frontend** with **TypeScript** and **reusable UI component library** for agent status visualization, invocation controls, and **error handling**; implemented **Jest unit tests (92% coverage)** and **Cypress e2e tests** for critical user flows, achieving **zero critical production incidents** over **6 months** of deployment.
- Engineered **prompt engineering** framework with **tool schema definitions**, **function calling patterns**, and **context window optimization** using **Python FastAPI**; implemented **RAG pipeline** with vector embeddings for codebase augmentation, reducing agent hallucination by **38%** and improving suggestion relevance from **62% to 89%** in user acceptance testing.
- Implemented **observability stack** with **OpenTelemetry**, structured JSON logging, and custom metrics dashboards for agent execution traces, latency analysis, and failure attribution; enabled **5+ product teams** to self-serve diagnostics and iterate on prompts without backend support, reducing **mean time to resolution by 45%**.
- Documented **OpenAI specification contracts**, **MCP protocol implementation guides**, **prompt versioning best practices**, and **evaluation frameworks** with **golden test sets** for model validation; created **internal SDK** and tutorial library enabling **15+ engineering teams** to ship agentic features in **2-week sprint cycles**.

SWIFT Payments Operations Dashboard – Real-Time Transaction Monitoring & ISO 20022 Migration Suite (HCL Technologies)

- Built and maintained production **payments operations dashboard** for SWIFT messaging and cross-border payment flows, processing **50,000+ transactions daily** with **99.95% uptime SLA**; implemented **real-time monitoring** using **Splunk and CloudWatch** with **P1/P2 incident alerting**, **SEPA and ISO 20022 message validation**, and transaction-flow visibility for financial operations teams.
- Engineered **Python ETL pipelines** and **SQL schema designs** across **PostgreSQL, MySQL, Oracle** for payment reconciliation workflows, tokenization data flows, and GDPR-compliant audit trails; automated **nightly reporting jobs** with **100% data accuracy** and **zero regulatory compliance violations** over **3+ years** of operation.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)